

Tan Dai Ngo

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EDUCATION

University of Victoria | Master of Engineering, Applied Data Science (Co-op) *Victoria, BC | Sep 2025 - Dec 2027*

- **Coursework:** Optimization for ML, Artificial Intelligence, Data Mining, Information Theory. **GPA:** 8.78 / 9.00.

University of Chicago | Master of Science, Computer Science *Chicago, IL, USA | Sep 2020 - Mar 2022*

- **Coursework:** Algorithms, Distributed Systems, Databases, Applied Software Engineering, Networks.

University of Washington | Bachelor of Science, Economics *Seattle, WA, USA | Sep 2015 - Aug 2019*

- **Minors:** Informatics, Applied Mathematics. **Certificate:** Quantitative Managerial Economics. **GPA:** 3.57 / 4.00.

EXPERIENCE

Software Developer | T-Mobile (via BeaconFire Inc.) *Bellevue, WA, USA | Jun 2022 - Apr 2024*

- Built and maintained REST endpoints across 5+ Roaming Business System microservices in Java and Spring Boot, keeping Cassandra query response times under 3 seconds.
- Automated reprocessing of rejected roaming tests with weekly Kafka retries, cutting manual effort by 80%.
- Traced 20+ cross service defects to their root cause and deployed features via Jenkins across 50+ releases.
- Debugged production issues with Splunk and built a Mobileum email notifier over the Microsoft Graph API.

Math and Computer Science Teacher | APU American International Schools *Da Nang, Vietnam | Aug - Dec 2024*

- Taught Python, algorithms, object oriented design and software engineering fundamentals to 50+ students.

PROJECTS

Multi-Stage IoT Intrusion Detection | Python, PyTorch, XGBoost *UVic Capstone, team of 7 | May - Aug 2026*

- Compared six detection models under a 1% false positive budget, reaching 99.6% recall at 0.82% FPR.
- Designed the four condition holdout that exposed session leakage in the benchmark, then re-scored every model on an honest split where PR-AUC landed at 0.242.
- Scripted an integrity check confirming every engineered feature stayed causal and label free.

Ocean Data ML Platform with RAG | Python, Spark, Qdrant, FastAPI *Personal Project | Feb - Mar 2026*

- Ingested 10M+ NOAA and Ocean Networks Canada observations into raw, cleaned and curated Spark layers, validating each tier so nothing downstream read partial data.
- Served model inference and plain English retrieval behind FastAPI using embeddings, Qdrant and Ollama, containerized with Docker Compose, at hit@k 0.90 and term recall 0.85.
- Tuned XGBoost with Optuna and tracked runs in MLflow, benchmarked against persistence on air temperature to cut RMSE 48% at the 12 hour horizon, and shipped a Streamlit dashboard.

Fuel Blending Prediction Service | Python, CatBoost, FastAPI, AWS *Shell.ai Hackathon 2025 | Jul 2025*

- Benchmarked CatBoost against XGBoost under 5-fold cross validation, selecting CatBoost at MAPE 0.64 against 1.29.
- Engineered entropy based mixture metrics and weighted aggregation across ten chemical blend properties.
- Deployed the model behind FastAPI and Docker on AWS EC2 for real time inference, serializing preprocessing so production predictions matched training.

TECHNICAL SKILLS

Languages: Python, Java, Go, SQL, JavaScript, MATLAB, Cypher, HTML & CSS

Machine Learning & AI: PyTorch, XGBoost, scikit-learn, pandas, NumPy, DistilBERT, Optuna, MLflow, RAG, Ollama

Data & Backend: Spark, PySpark, ETL pipelines, Kafka, Cassandra, MySQL, MongoDB, Qdrant, FastAPI, Spring Boot, REST

Platform & Tools: Docker, AWS (EC2, S3), Jenkins CI/CD, Git, Splunk, Streamlit, Jira, Confluence

Certifications & Awards: Google Advanced Data Analytics Specialization (Jun 2025); IMC 64 Bids, top 15% of 1,500